torsi'on

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Problema 1

$$\sigma_x = -840 \text{ kg/m}^2, \quad \sigma_y = 1050 \text{ kg/m}^2, \quad \tau_{xy} = 560 \text{ kg/m}^2$$

1. Esfuerzos principales ($\sigma_{\mathbf{max}}, \sigma_{\mathbf{min}}$)

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{\text{prom}} = \frac{-840 + 1050}{2} = 105$$

$$R = \sqrt{\left(\frac{-840 - 1050}{2}\right)^2 + 560^2} = \sqrt{(-945)^2 + 560^2}$$

$$R = \sqrt{893025 + 313600} = \sqrt{1206625} \approx 1098.465$$

Entonces:

$$\sigma_1 = 105 + 1098.465 \approx 1203.465 \text{ kg/m}^2$$

$$\sigma_2 = 105 - 1098.465 \approx -993.465 \text{ kg/m}^2$$

$$\boxed{\sigma_{\max} = 1203.47 \text{ kg/m}^2}$$

$$\boxed{\sigma_{\min} = -993.47 \text{ kg/m}^2}$$

2. Ángulo de esfuerzos principales ($2\theta_p$)

$$\tan(2\theta_p) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{1120}{-1890} \approx -0.59259$$
$$2\theta_p = \arctan(-0.59259) \approx -30.64^\circ$$

$$\boxed{2\theta_p \approx -30.64^\circ}$$

3. Esfuerzo cortante máximo ($\tau_{\mathbf{max}}$)

$$\tau_{\max} = R = 1098.465 \text{ kg/m}^2$$
$$\boxed{\tau_{\max} = 1098.47 \text{ kg/m}^2}$$

4. Esfuerzo normal en el plano de $\tau_{\mathbf{max}}$

$$\sigma_n = \frac{\sigma_x + \sigma_y}{2} = 105 \text{ kg/m}^2$$
$$\boxed{\sigma_n = 105 \text{ kg/m}^2}$$

5. Ángulo para cortante máximo ($2\theta_c$)

$$2\theta_c = 2\theta_p + 90^\circ \approx -30.64^\circ + 90^\circ = 59.36^\circ$$
$$\boxed{2\theta_c \approx 59.36^\circ}$$

6. Para $\theta = 20^\circ$, hallar σ, τ

Fórmulas de transformación:

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$
$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

Para $\theta = 20^\circ \rightarrow 2\theta = 40^\circ$:

$$\sigma = 105 + \left(\frac{-840 - 1050}{2}\right) \cos 40^\circ + 560 \sin 40^\circ$$
$$\sigma = 105 + (-945) \times 0.766044 + 560 \times 0.642788$$

$$\sigma = 105 - 723.912 + 359.961 \approx -258.951 \text{ kg/m}^2$$

$$\tau = -\left(\frac{-840 - 1050}{2}\right) \sin 40^\circ + 560 \cos 40^\circ$$

$$\tau = -(-945) \times 0.642788 + 560 \times 0.766044$$

$$\tau = 607.434 + 428.985 \approx 1036.419 \text{ kg/m}^2$$

$$\boxed{\sigma \approx -258.95 \text{ kg/m}^2, \quad \tau \approx 1036.42 \text{ kg/m}^2}$$

Resumen final:

1. $\sigma_{\max} \approx 1203.47 \text{ kg/m}^2$
2. $\sigma_{\min} \approx -993.47 \text{ kg/m}^2$
3. $2\theta_p \approx -30.64^\circ$
4. $\tau_{\max} \approx 1098.47 \text{ kg/m}^2$
5. $\sigma_n = 105 \text{ kg/m}^2$
6. $2\theta_c \approx 59.36^\circ$
7. Para $\theta = 20^\circ$: $\sigma \approx -258.95 \text{ kg/m}^2, \ \tau \approx 1036.42 \text{ kg/m}^2$

Problema 2

$$\sigma_x = 840 \text{ kg/m}^2, \quad \sigma_y = 1050 \text{ kg/m}^2, \quad \tau_{xy} = -560 \text{ kg/m}^2$$

1. Esfuerzos principales ($\sigma_{\max}, \sigma_{\min}$)

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{\text{prom}} = \frac{840 + 1050}{2} = 945$$

$$R = \sqrt{\left(\frac{840 - 1050}{2}\right)^2 + (-560)^2} = \sqrt{(-105)^2 + 313600}$$

$$R = \sqrt{11025 + 313600} = \sqrt{324625} \approx 569.76$$

Entonces:

$$\sigma_1 = 945 + 569.76 \approx 1514.76 \text{ kg/m}^2$$

$$\sigma_2 = 945 - 569.76 \approx 375.24 \text{ kg/m}^2$$

$$\boxed{\sigma_{\max} = 1514.76 \text{ kg/m}^2}$$

$$\boxed{\sigma_{\min} = 375.24 \text{ kg/m}^2}$$

2. Ángulo de esfuerzos principales ($2\theta_p$)

$$\tan(2\theta_p) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{-1120}{-210} \approx 5.3333$$
$$2\theta_p = \arctan(5.3333) \approx 79.38^\circ$$

$$\boxed{2\theta_p \approx 79.38^\circ}$$

3. Esfuerzo cortante máximo ($\tau_{\max}$)

$$\tau_{\max} = R = 569.76 \text{ kg/m}^2$$
$$\boxed{\tau_{\max} = 569.76 \text{ kg/m}^2}$$

4. Esfuerzo normal en el plano de $\tau_{\max}$

$$\sigma_n = \frac{\sigma_x + \sigma_y}{2} = 945 \text{ kg/m}^2$$
$$\boxed{\sigma_n = 945 \text{ kg/m}^2}$$

5. Ángulo para cortante máximo ($2\theta_c$)

$$2\theta_c = 2\theta_p + 90^\circ \approx 79.38^\circ + 90^\circ = 169.38^\circ$$
$$\boxed{2\theta_c \approx 169.38^\circ}$$

6. Para $\theta = 20^\circ$, hallar σ, τ

Fórmulas de transformación:

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$
$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$$

Para $\theta = 20^\circ \to 2\theta = 40^\circ$:

$$\sigma = 945 + \left(\frac{840 - 1050}{2}\right)\cos 40^\circ + (-560)\sin 40^\circ$$
$$\sigma = 945 + (-105) \times 0.766044 - 560 \times 0.642788$$
$$\sigma = 945 - 80.435 - 359.961 \approx 504.604 \text{ kg/m}^2$$

$$\tau = -\left(\frac{840 - 1050}{2}\right)\sin 40^\circ + (-560)\cos 40^\circ$$
$$\tau = -(-105) \times 0.642788 - 560 \times 0.766044$$
$$\tau = 67.493 - 428.985 \approx -361.492 \text{ kg/m}^2$$

$$\boxed{\sigma \approx 504.60 \text{ kg/m}^2, \quad \tau \approx -361.49 \text{ kg/m}^2}$$

Resumen final:

1. $\sigma_{\max} \approx 1514.76 \text{ kg/m}^2$
2. $\sigma_{\min} \approx 375.24 \text{ kg/m}^2$
3. $2\theta_p \approx 79.38^\circ$
4. $\tau_{\max} \approx 569.76 \text{ kg/m}^2$
5. $\sigma_n = 945 \text{ kg/m}^2$
6. $2\theta_c \approx 169.38^\circ$
7. Para $\theta = 20^\circ$: $\sigma \approx 504.60 \text{ kg/m}^2$, $\tau \approx -361.49 \text{ kg/m}^2$

Problema 4

$$\sigma_x = -840 \text{ kg/m}^2, \quad \sigma_y = 0 \text{ kg/m}^2, \quad \tau_{xy} = 280 \text{ kg/m}^2$$

1. Esfuerzos principales ($\sigma_{\max}, \sigma_{\min}$)

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{\text{prom}} = \frac{-840 + 0}{2} = -420$$
$$R = \sqrt{\left(\frac{-840 - 0}{2}\right)^2 + 280^2} = \sqrt{(-420)^2 + 280^2}$$
$$R = \sqrt{176400 + 78400} = \sqrt{254800} \approx 504.78$$

Entonces:

$$\sigma_1 = -420 + 504.78 \approx 84.78 \text{ kg/m}^2$$
$$\sigma_2 = -420 - 504.78 \approx -924.78 \text{ kg/m}^2$$

$$\boxed{\sigma_{\max} = 84.78 \text{ kg/m}^2}$$
$$\boxed{\sigma_{\min} = -924.78 \text{ kg/m}^2}$$

2. Ángulo de esfuerzos principales ($2\theta_p$)

$$\tan(2\theta_p) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{560}{-840} \approx -0.66667$$
$$2\theta_p = \arctan(-0.66667) \approx -33.69^\circ$$

$$\boxed{2\theta_p \approx -33.69^\circ}$$

3. Esfuerzo cortante máximo ($\tau_{\max}$)

$$\tau_{\max} = R = 504.78 \text{ kg/m}^2$$
$$\boxed{\tau_{\max} = 504.78 \text{ kg/m}^2}$$

4. Esfuerzo normal en el plano de $\tau_{\max}$

$$\sigma_n = \frac{\sigma_x + \sigma_y}{2} = -420 \text{ kg/m}^2$$
$$\boxed{\sigma_n = -420 \text{ kg/m}^2}$$

5. Ángulo para cortante máximo ($2\theta_c$)

$$2\theta_c = 2\theta_p + 90^\circ \approx -33.69^\circ + 90^\circ = 56.31^\circ$$
$$\boxed{2\theta_c \approx 56.31^\circ}$$

6. Para $\theta = 20^\circ$, hallar σ, τ

Fórmulas de transformación:

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$
$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$$

Para $\theta = 20^\circ \rightarrow 2\theta = 40^\circ$:

$$\sigma = -420 + \left(\frac{-840 - 0}{2}\right)\cos 40^\circ + 280\sin 40^\circ$$
$$\sigma = -420 + (-420) \times 0.766044 + 280 \times 0.642788$$
$$\sigma = -420 - 321.738 + 179.981 \approx -561.757 \text{ kg/m}^2$$

$$\tau = -\left(\frac{-840 - 0}{2}\right)\sin 40^\circ + 280\cos 40^\circ$$
$$\tau = -(-420) \times 0.642788 + 280 \times 0.766044$$
$$\tau = 269.971 + 214.492 \approx 484.463 \text{ kg/m}^2$$

$$\boxed{\sigma \approx -561.76 \text{ kg/m}^2, \quad \tau \approx 484.46 \text{ kg/m}^2}$$

Resumen final:

1. $\sigma_{\max} \approx 84.78$ kg/m^2
2. $\sigma_{\min} \approx -924.78$ kg/m^2
3. $2\theta_p \approx -33.69^\circ$
4. $\tau_{\max} \approx 504.78$ kg/m^2
5. $\sigma_n = -420$ kg/m^2
6. $2\theta_c \approx 56.31^\circ$
7. Para $\theta = 20^\circ$: $\sigma \approx -561.76$ kg/m^2, $\tau \approx 484.46$ kg/m^2